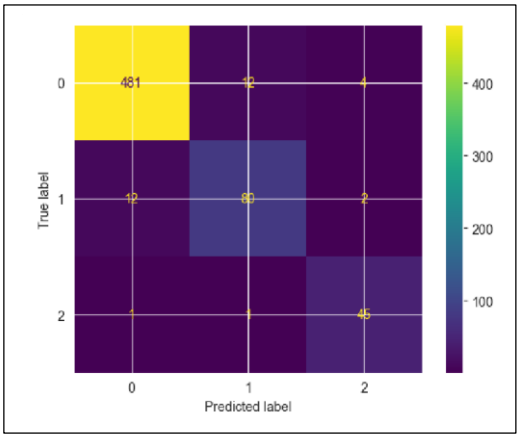
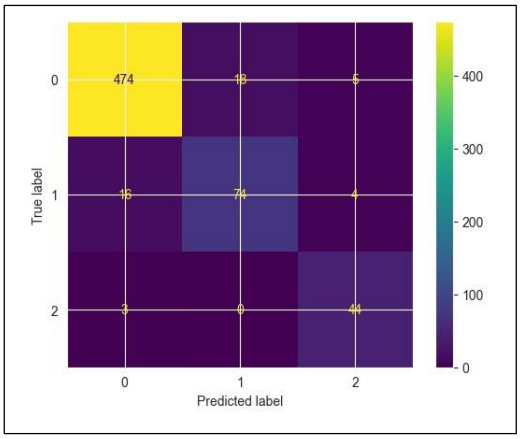
Project Development Phase
Model Performance Test

Model Performance Testing:

S.No.	Parameter	Values	Screenshot
1.	Metrics	**Classification Model:** **a) Random Forest** Confusion Matrix Accuracy Score	 |	Name	Score
---	---	---
0	Random Forest Classifier	0.949843
1	Decision Tree Classifier	0.927900
2	Logistic Regression	0.810345
3	K Neighbors Classifier	0.898119	

b) Decision Tree

Confusion Matrix

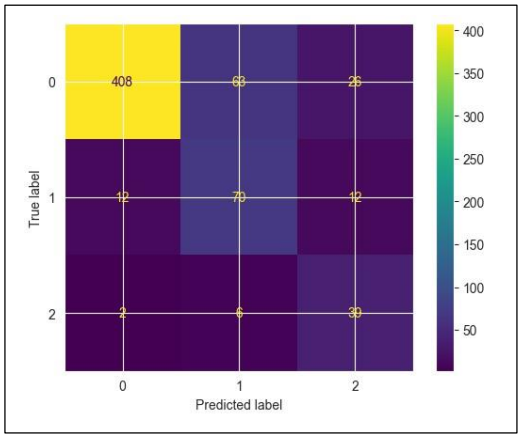


Accuracy Score

	Name	Score
0	Random Forest Classifier	0.949843
1	Decision Tree Classifier	0.927900
2	Logistic Regression	0.810345
3	K Neighbors Classifier	0.898119

c) Logistic Regression

Confusion Matrix



Accuracy Score

	Name	Score
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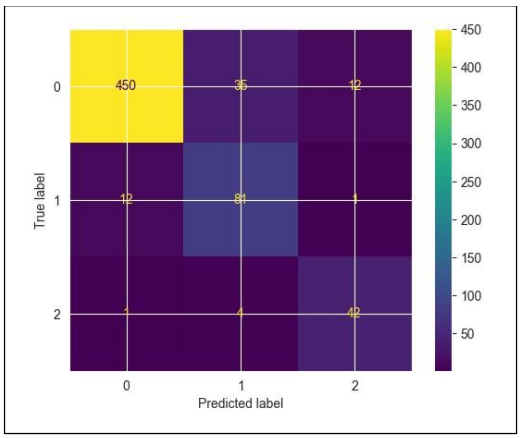
3	K Neighbors Classifier	0.898119

d) K-Nearest Neighbours

Confusion Matrix



Accuracy Score

	Name	Score
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2.	Tune the Model	Validation Method	

```python
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,y, test_size=0.3, random_state=10)

X_train.shape, X_test.shape
```

```python
RF_model.predict([[0.001,26,0,41,132,8.2,133,0.005]])
```

```
array([1.])
```

```python
RF_model.predict([[0.001,34,0,116,131,12.4,99,0.002]])
```

```
array([2.])
```

```python
RF_model.predict([[0,87,71.0,0,125,3.4,124,0]])
```

```
array([3.])
```